Chapter 4

Gas and Solution

Learning Objectives

- Define ideal and real gases
- Explain the ideal gas law, Boyle’s Law, Charles’ law, Avogadro’s law and Dalton’s law
- Explain the limitations of ideality at very high pressures and very low temperatures
- Describe the types of solutions, solution process relating to intermolecular forces and hydrogen bond
- Explain solubility in terms of units of concentration
- Understand the colligative properties of solution
- Understand colloid

Part 1

Gases and Gas Laws

Gases

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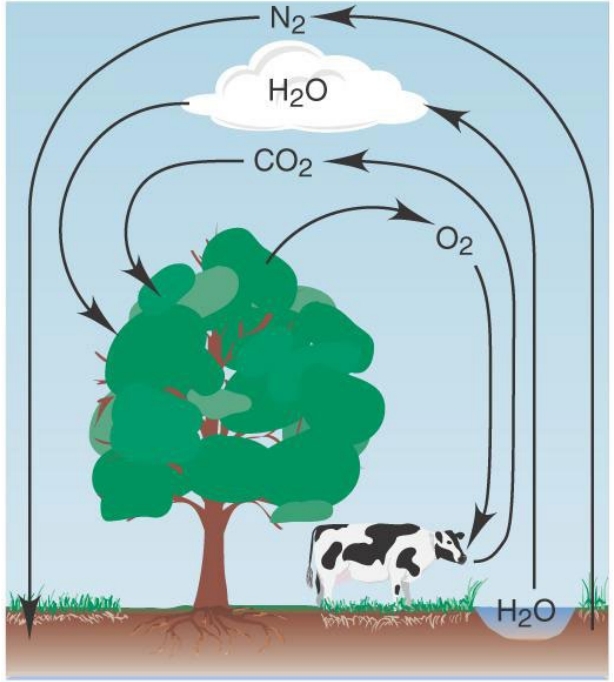

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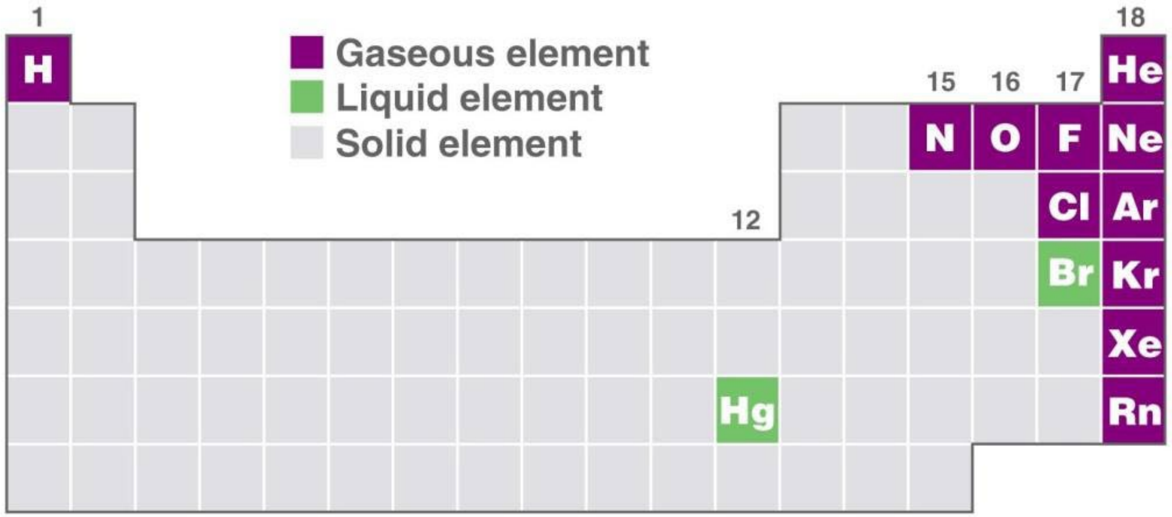
Gaseous Elements

Common gaseous compounds



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- Elements that occur as gases are on the right side of the periodic table except for hydrogen
- All the noble gases (Group 18) are monatomic gases
- The other gaseous elements are diatomic molecules (H_2, N_2, O_2, F_2, and Cl_2)

Distinction of gases from liquids and solids

1. Gas volume changes greatly with pressure.
2. Gas volume changes greatly with temperature.
3. Gases have relatively low viscosity.
4. Most gases have relatively low densities under normal conditions.
5. Different gases are always miscible. [Not all liquids are miscible.]

The physical behavior of gases

- Described by four variables:
  - Pressure: p
  - Volume: V
  - Temperature: T
  - Amount of the gas (number of moles): n
- The variables are interdependent → Any one of them can be determined by measuring the other three.

- The three key relationships among the four gas variables are
  – Boyle's law (p-V)
  – Charles' law (V-T)
  – Avogadro's law (V-n)

Gay-Lussac's Law (p-T)
Dalton's Law of partial pressure

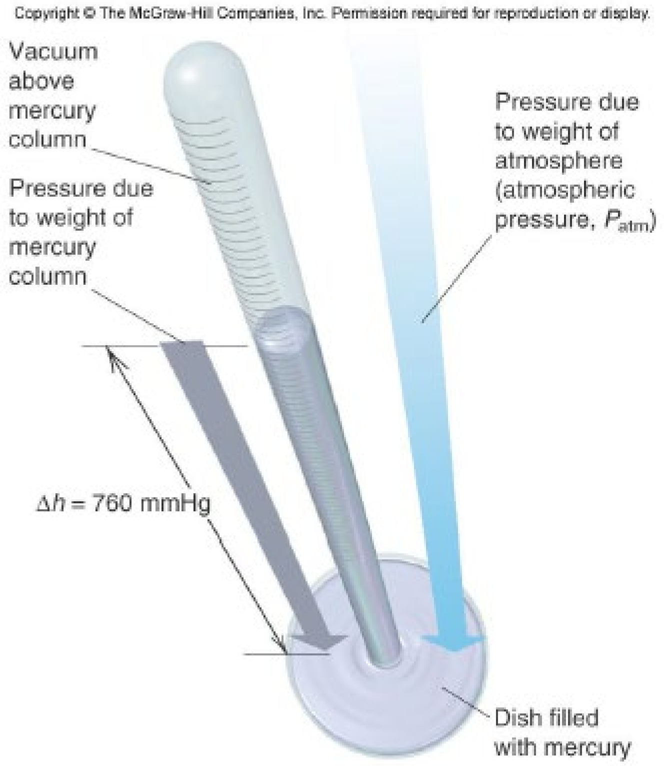
Gas Pressure

$$P = F/A$$

- Pressure is the force exerted per unit area.

- Every point on Earth's surface experiences a net pressure called ***atmospheric pressure.***

- Atmospheric pressure can be measured using a **barometer** - a closed, inverted tube filled with mercury.



- **Standard atmospheric pressure is referred to as 1 *atmosphere (atm)***
- ***1 atm = 760 mmHg = 760 torr = 1.01325×10^5 Pa = 101.325 kPa***

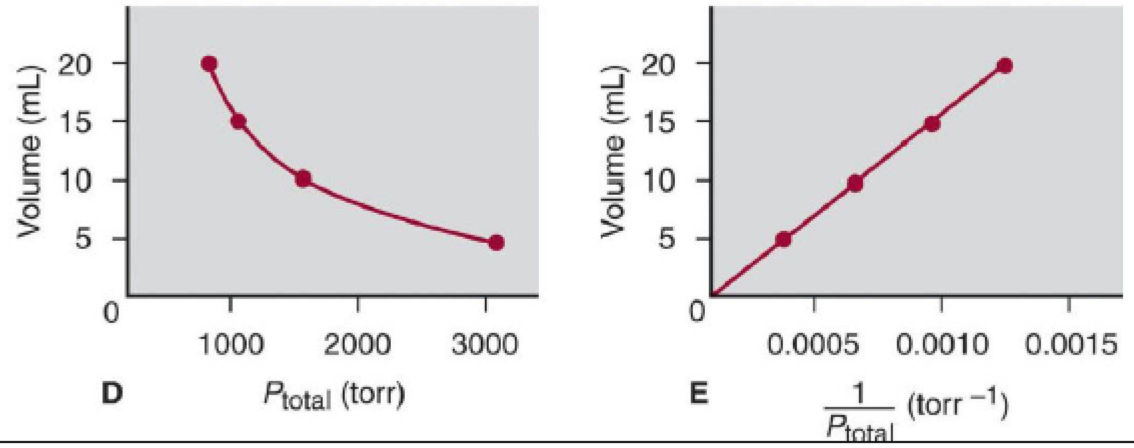
Boyle’s Law – pressure vs volume

- English Chemist Robert Boyle concluded that at a given temperature the volume occupied by the gas is inversely related to its pressure.
- Plots of Boyle’s data showed that a simple plot of *V versus P* is a hyperbola (双曲线)

$$V \propto \frac{1}{P}$$ [n and T are fixed] ***PV* = constant.**

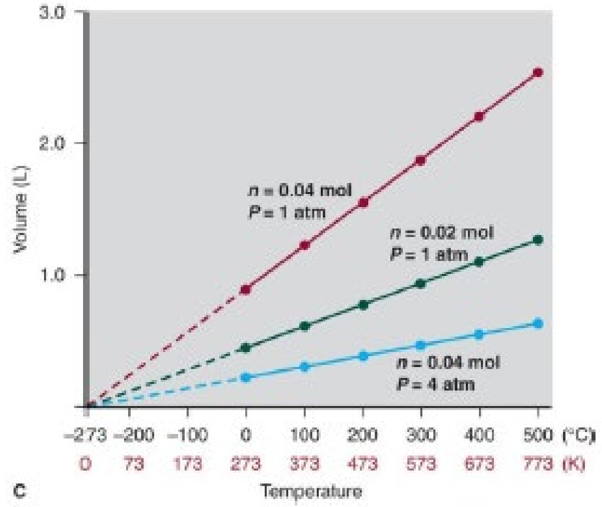
Numerical value of the constant depends on the amount of gas and the temperature.

A plot of ***V* versus 1/*P*** is a straight line whose slope is equal to the constant.



Charles' Law – volume vs temperature

- At constant pressure the volume occupied by a fixed amount of gas is directly proportional to its absolute temperature (Kelvin)



$$V \propto T \qquad \text{[P and n are fixed]}$$

- Two other relationships in gas behavior

  that emerge from Boyle's and Charles' laws are given below :

1. At constant volume, the pressure exerted by a fixed amount of gas is directly proportional to the absolute temperature

$$P \propto T \qquad \text{[V and n are fixed]}$$

2. A simple combination of Boyle's and Charles' law gives the combined gas law

$$\frac{PV}{T} = \text{constant}$$

Avogadro's law – volume vs amount

- At fixed temperature and pressure, the volume occupied by a gas is directly proportional to the amount (mol) of gas

$V \propto n$ (P and T fixed)

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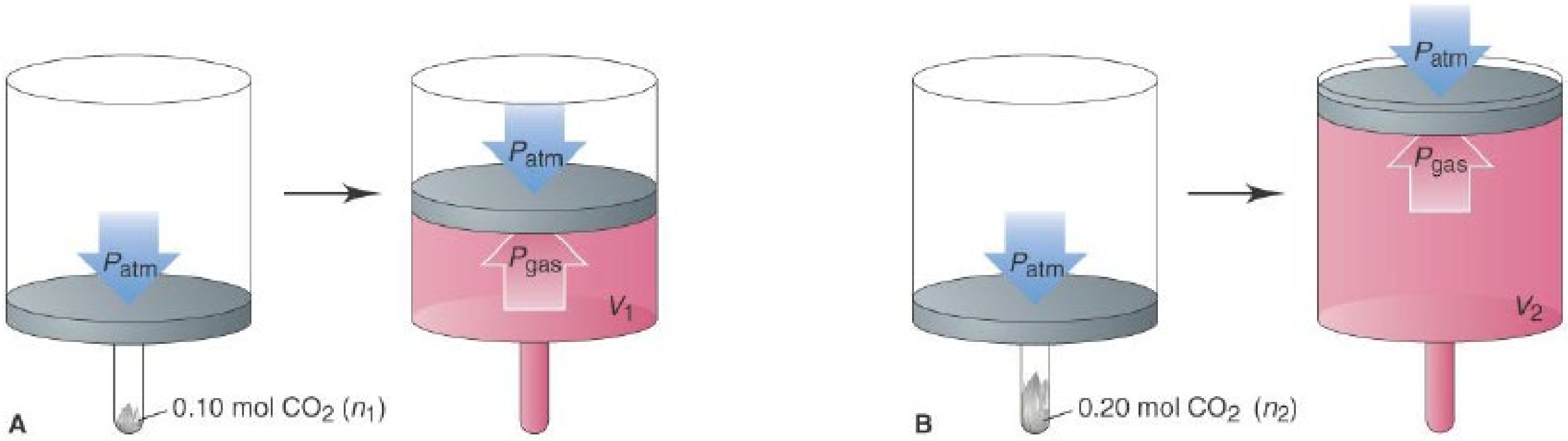


Deriving the Ideal Gas Law

- Combine the following three expressions

$$V \propto 1/P \text{ (at constant } n, T)$$
$$V \propto T \text{ (at constant } n, P)$$
$$V \propto n \text{ (at constant } T, P)$$

We get

$$V \propto nT/P \quad \text{or} \quad V = \text{constant } (nT/P)$$

- The proportionality constant is called the **gas constant, *R*.**

(R= 8.314 J mol^{-1} K^{-1})

- ***Inserting R into the equation gives*** $V = \dfrac{RnT}{P} = \dfrac{nRT}{P}$

$$\boxed{PV = nRT}$$

Application of Ideal Gas Law

$PV = nRT$

- Molar volume of gases

- Density of a gas

- Molar mass of a gas

Partial Pressures

- The **total pressure** exerted by a mixture of gases at a given temperature and volume is the **sum** of the pressures exerted by each of the gases alone. These individual gas pressures are called **partial pressures**.

$$\mathbf{P_t = P_1 + P_2 + P_3 \ldots + P_i}$$

- Partial pressure is the pressure the gas **would** exert **if** it were the only one present (at the same temperature and volume).
- Air is ____% N_2 and ____% O_2. CO_2 in air is 420 ppm. Calculate the partial pressure of each in Pa.

Mole Fractions of Gas Mixtures

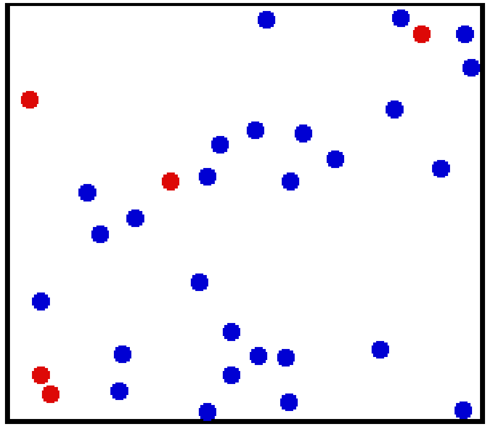
- The partial pressure of any gas in a mixture is the total pressure multiplied by the mole fraction of that gas.

$P_A = \chi_A \times P_{total}$ where χ_A is the mole fraction

$$\chi_1 = \frac{n_A}{n_A + n_B + n_C + \ldots} = \frac{n_1}{n_{total}}$$

Χ χ	chi, χι	[kʰ]	c as in English *cat*[15][note 2]	[x] before [a], [o], [u]; [ç] before [e], [i]	ch as in Scottish *loch*; h as in English *hue*

The kinetic theory of gases



- Kinetic molecular theory of gases provides a molecular-level explanation for the observed properties of gases

Postulate 1: *Particle Volume*

- The volume of an individual gas particle is so small compared to the volume of its container → the gas particles have no volume (accurately speaking, negligible volume).

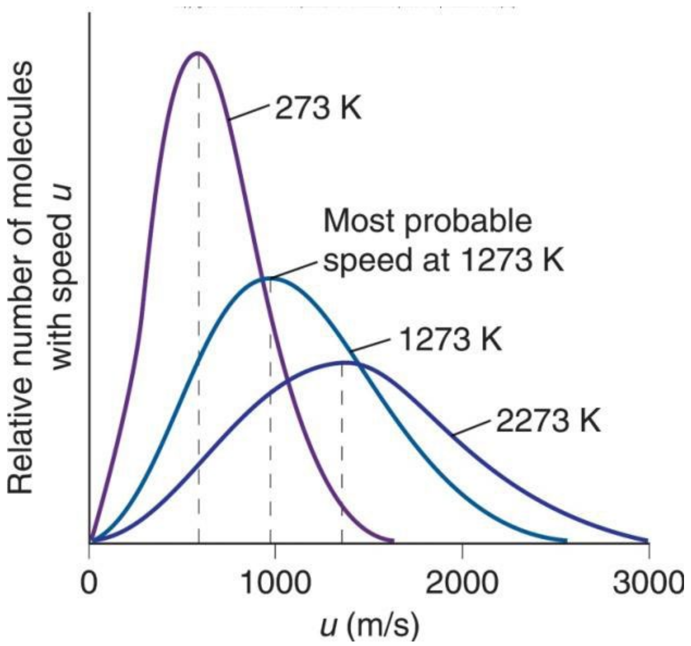
Postulate 2: *Particle Motion*

- Gas particles are in constant, random, straight-line motion except when they collide with each other or with the container walls.

Postulate 3: *Particle Collisions*

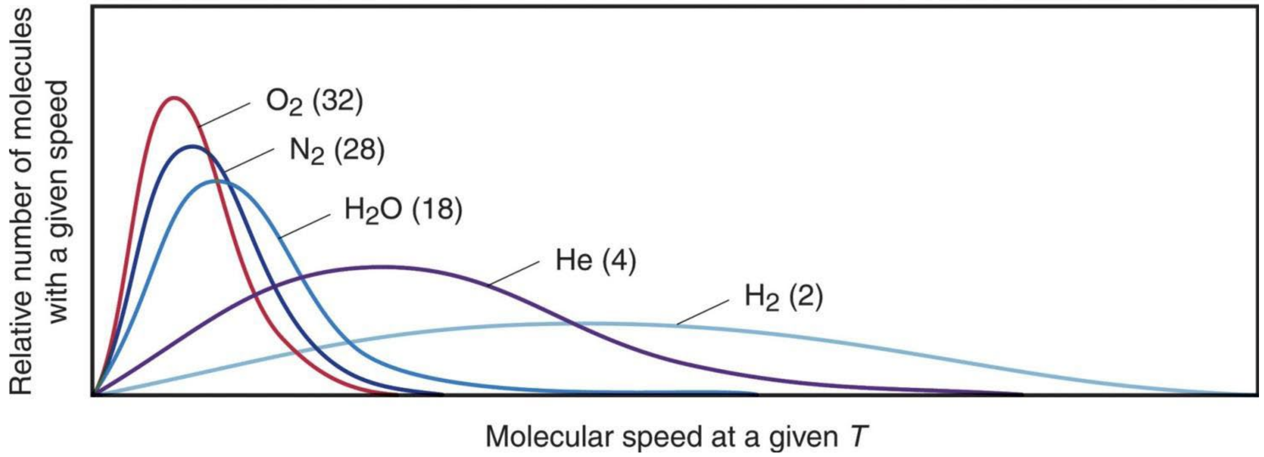
- Collisions are elastic, therefore, the total kinetic energy (E_k) of all the particles is constant.

Maxwell–Boltzmann distribution



- Curve is not symmetrical
- When T increases:
  - The most probable speed increases
  - The percentage of molecule with high speed (e.g. 2000 m/s) increases

This is the basis of the Arrhenius equation of kinetics



At a given temperature, gas with lower molar mass has higher speed (most probable speed).

Real Gases: Deviation from Ideal behavior

- In reality, molecules have volumes.
- Also, molecules have attractive / repulsive interactions (due to their dipole moment) even when there is no collision.
- Hence the real gases deviate from ideal behavior.

- In 1873 Johannes van der Waals proposed an equation for 'n' moles of a real gas.

The deviation is more significant at very high pressure and low temperature

High p → more molecules per unit volume → molecules are closer → attractive/repulsive interactions are stronger

Low T → KE is smaller → potential energy contribution is relatively more

Van der Waals equation for n moles of a real gas

$$(P + \frac{n^2 a}{V^2})(V - nb) = nRT$$

P=measured pressure, V=Container volume, n=number of moles of the gas
T=Temperature, a,b= Vander Waals constants

	a atm L^2 / mol^2	b L / mol
H_2	0.244	0.0266
He	0.0341	0.0237
N_2	1.39	0.0391
O_2	1.36	0.0318
Cl_2	6.49	0.0562
NH_3	4.17	0.0371
CO	1.49	0.0399
CO_2	3.59	0.0427

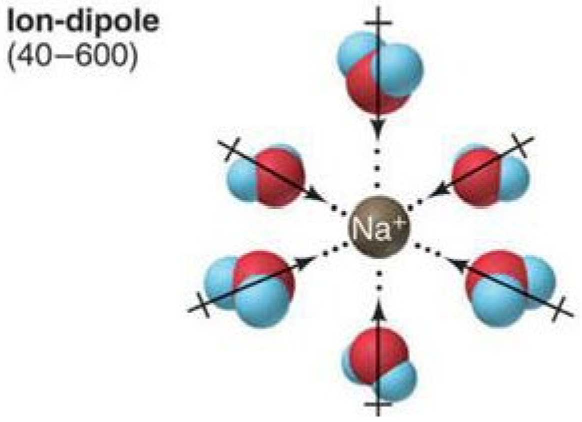
Part 2

Solution and Colloid

Solution

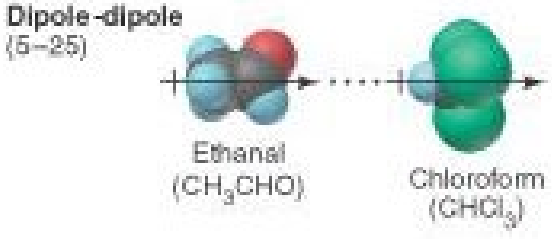
- A solution is a homogeneous mixture of two or more substances.
- A solution consists of a *solvent and one or more solutes.*
- In all solutions, whether gaseous, liquid, or solid, the substance present in the greatest amount is the *solvent, while the substance or substances present in smaller amounts are the solute(s).*
- Solutes do not have to be in the same physical state as the solvent but the physical state of the solvent determines the state of the solution.
- If the solute and solvent combine to give a homogeneous solution, the solute is said to be *soluble in the solvent.*

Intermolecular forces in solutions



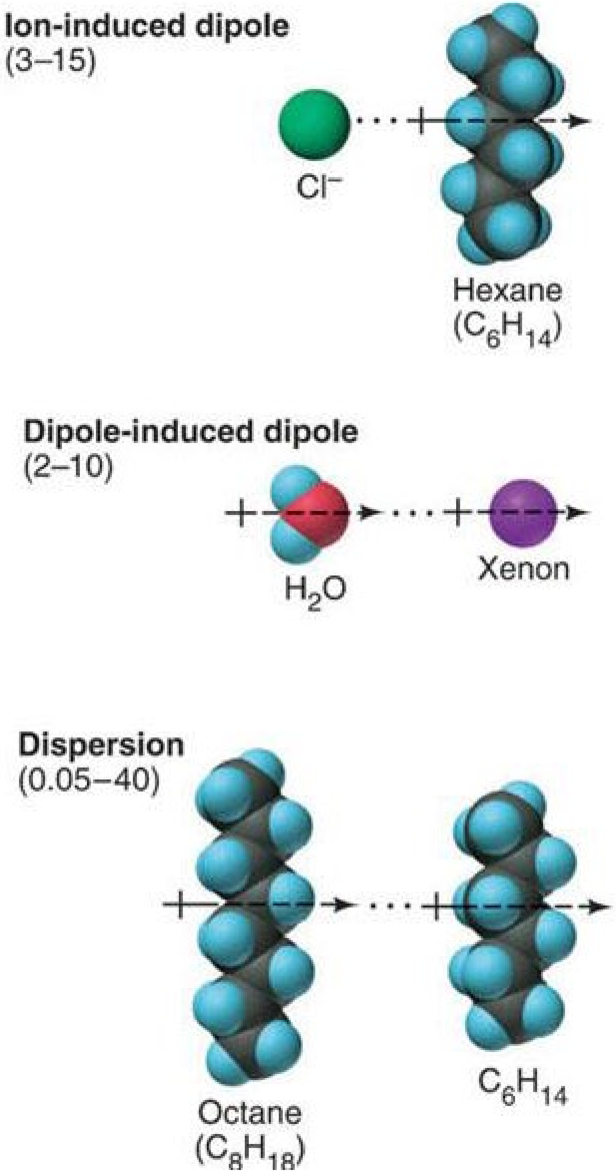
- Solubility of the ionic compounds in water involves ion-dipole forces. The ion on the crystal's surface attracts the end of the water molecule dipole that is opposite in charge.

- Hydrogen bonding is a primary factor in aqueous solutions.



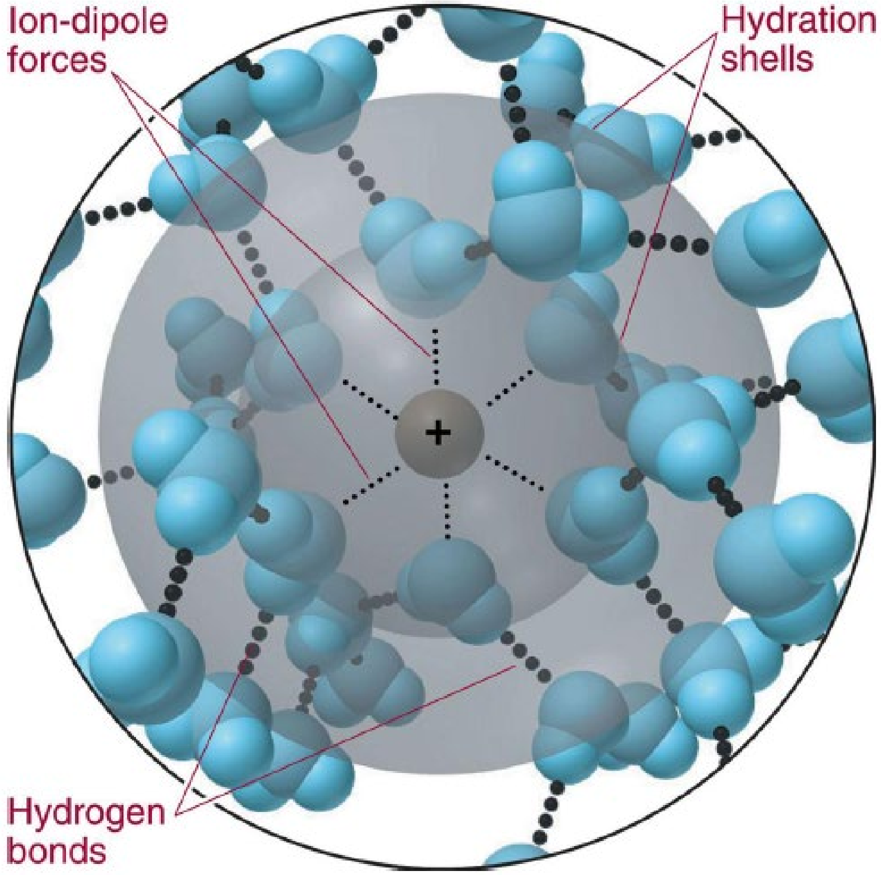
- Dipole-dipole forces account for the solubility of polar organic molecules in polar non-aqueous solvents (e.g. $CHCl_3$).

Intermolecular forces in solutions



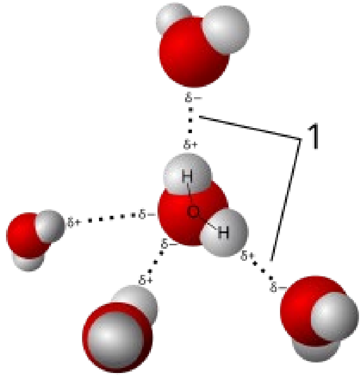
- Ion-induced dipole forces result when an ion's charge distorts the electron cloud of a nearby non-polar molecule.
  - Strong or weak?
- Dipole-induced dipole forces are weaker than the ion-induced dipole forces because the magnitude of the charge is smaller. This arises when a polar molecule distorts the electron cloud of a nearby non-polar molecule
- Dispersion forces contribute to the solubility of all solutes in all solvents. They are the principal forces in the solutions of non-polar substances.

Hydration shells around an aqueous ion



Hydrogen Bond

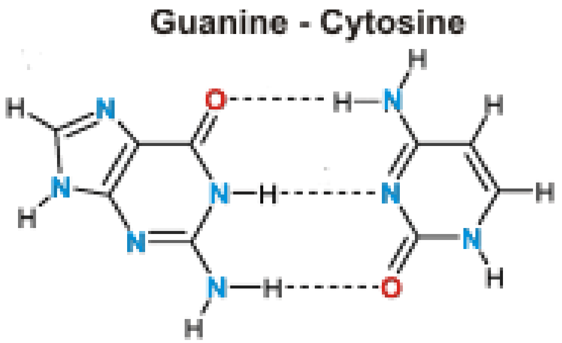
Water

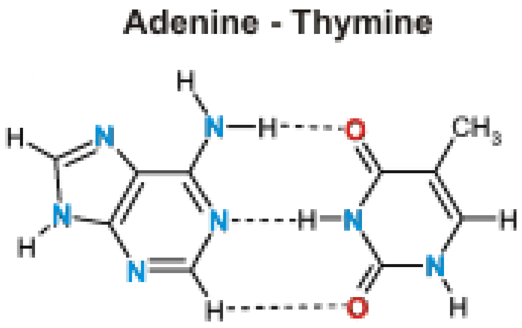


- Hydrogen bonding
- X – H ··· Y
- X and Y can be the same element; X and Y can also be different elements.
- X and Y must be one of the three elements: F, O, N

Aqueous ammonia

DNA





“Like dissolves like”

- The forces created between solute and solvent must be comparable in strength to the forces destroyed within pure solute and pure solvent.

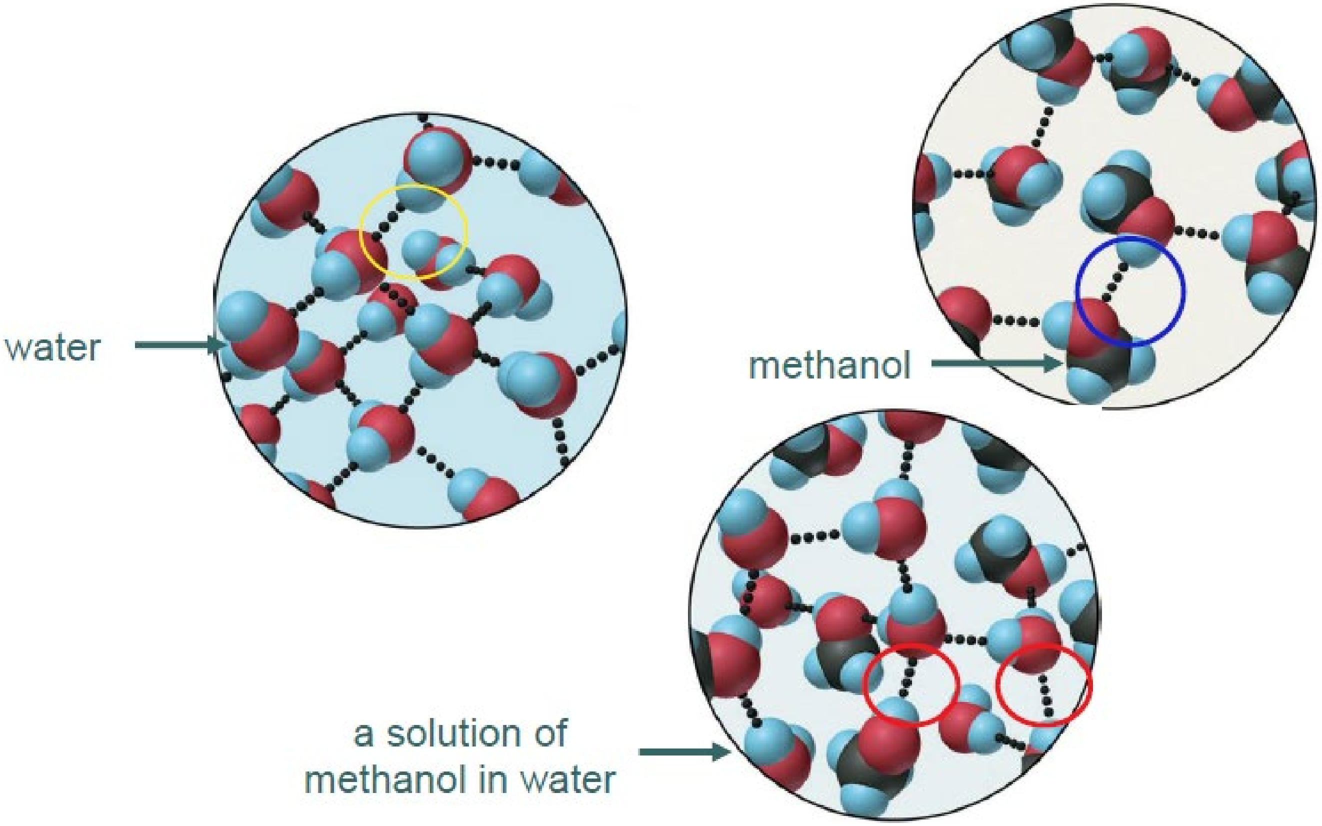


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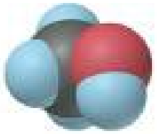
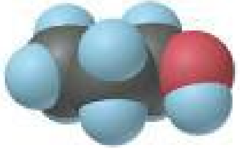
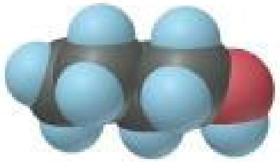
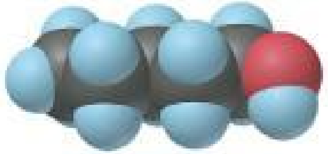
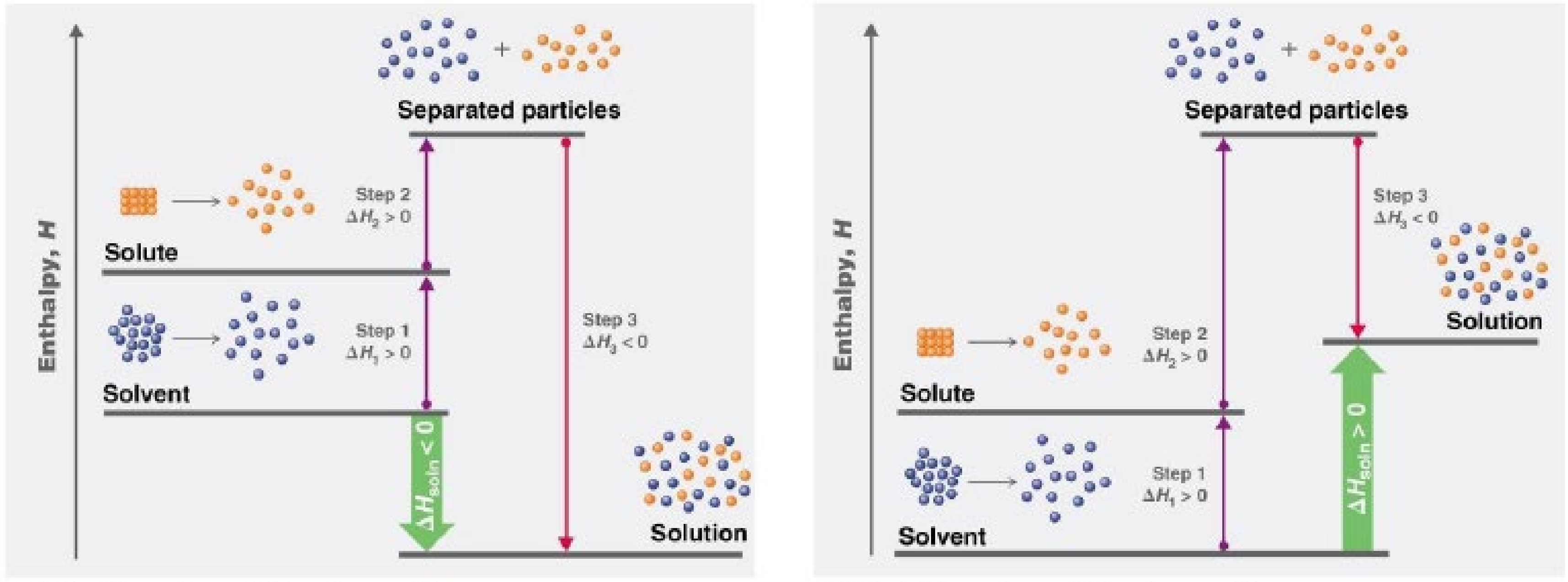
Table 13.2 Solubility* of a Series of Alcohols in Water and Hexane

Alcohol	Model	Solubility in Water	Solubility in Hexane
CH_3OH (methanol)		∞	1.2
CH_3CH_2OH (ethanol)		∞	∞
$CH_3(CH_2)_2OH$ (1-propanol)		∞	∞
$CH_3(CH_2)_3OH$ (1-butanol)		1.1	∞
$CH_3(CH_2)_4OH$ (1-pentanol)		0.30	∞
$CH_3(CH_2)_5OH$ (1-hexanol)		0.058	∞

*Expressed in mol alcohol/1000 g solvent at 20°C.

Dissolution Process

- Formation of a solution from a solute and a solvent is a physical process, not a chemical process, and does not involve a chemical transformation, but involves energy changes.



(a) Exothermic solution formation

(b) Endothermic solution formation

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Step 1: Separation of solute particles

$\Delta H_{solute} > 0$

Step 2: Separation of solvent particles

$\Delta H_{solute} > 0$

Step 3 Solute and solvent particles mix

$\Delta H_{mix} < 0$

$$\Delta H_{solution} = \Delta H_{solute} + \Delta H_{solute} + \Delta H_{mix}$$

Solubility

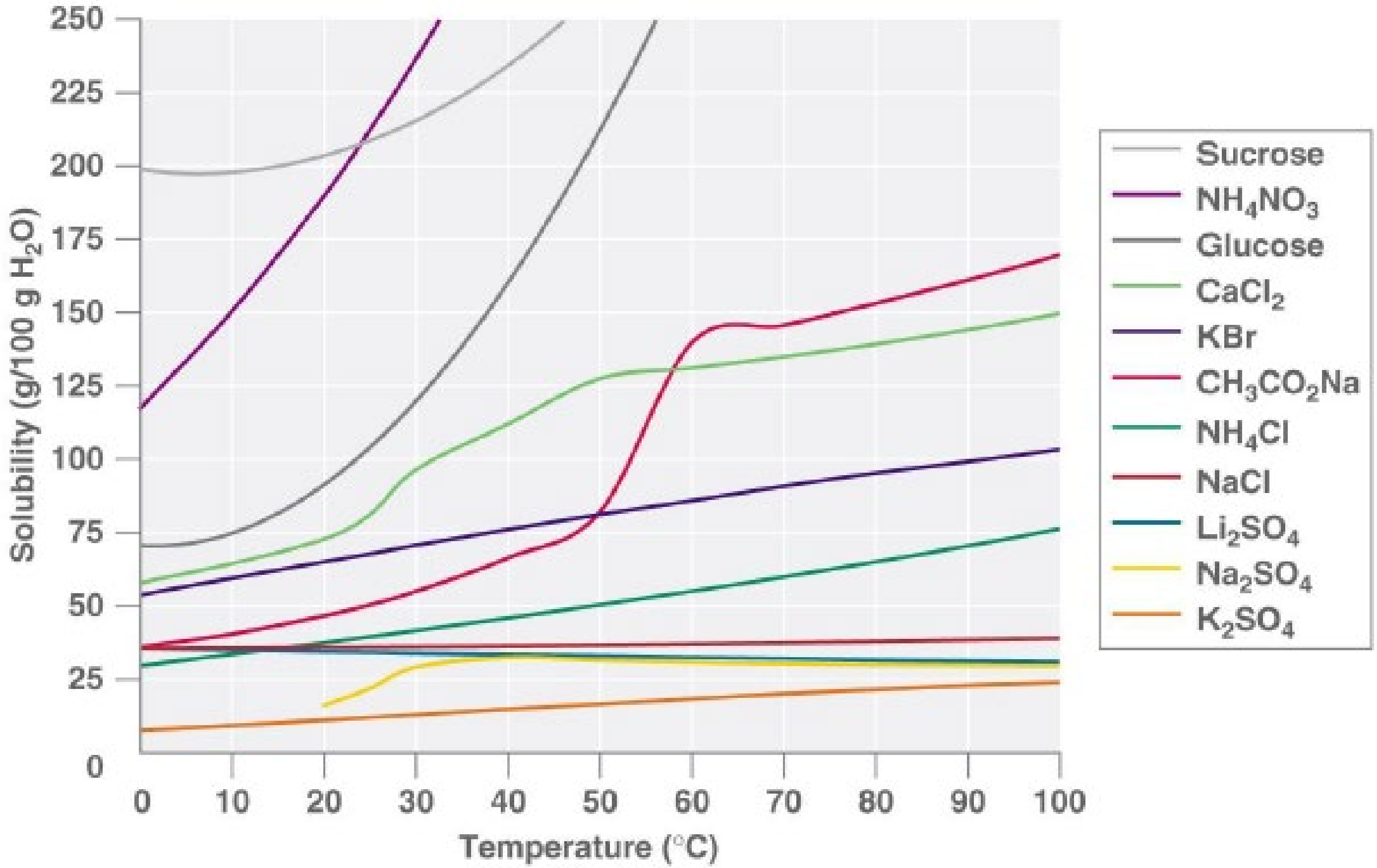
- Maximum amount of a solute that can dissolve in a solvent at a specified temperature and pressure is its **solubility.**
- Solubility of a substance depends on energetic factors and on the temperature and, for gases, the pressure.
- When an ionic solid is dissolved in a solvent, there is an equilibrium between the undissolved solute and dissolved solute

$$\text{Solute (undissolved)} \rightleftharpoons \text{Solute (dissolved)}$$

- Saturated solution: contains the maximum amount of dissolved solute at a given temperature
- Unsaturated solution: contains solute less than the maximum amount
- Supersaturated solution: contains solute more than the maximum amount → unstable

Solubility vs Temperature

- Solubility may increase or decrease with temperature, but generally it increases with increasing temperature [for solid].
- The magnitude of this temperature dependence varies widely among compounds.
- For gas, the solubility decreases with increasing temperature.



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Quantitative ways of expressing concentration

- Concentration is the proportion of a substance in a mixture. It is most often expressed as the ratio of the quantity of solute to the quantity of the solution/solvent.

Concentration Term	Ratio
Molarity (M)	$\frac{\text{amount (mol) of solute}}{\text{volume (L) of solution}}$
Molality (m)	$\frac{\text{amount (mol) of solute}}{\text{mass (kg) of solvent}}$
Mass percent	$\frac{\text{mass of solute X100}}{\text{mass of solute+mass of the solvent}}$
Volume percent	$\frac{\text{volume of solute X100}}{\text{volume of solution}}$
Mole fraction (χ)	$\frac{\text{amount (mol) of solute}}{\text{amount (mol) of solute + amount (mol) of solvent}}$
Mole percent (mol %)	$\frac{\text{amount (mol) of solute X100}}{\text{amount (mol) of solute + amount (mol) of solvent}}$

Colligative Properties

- Depend only on the ratio of the number of solute particles to the number of solvent particles in a solution, but not on the nature of the chemical species present.

- Colligative properties include:
  - Osmotic pressure
  - Elevation of boiling point
  - Depression of freezing point
  - Lowering of vapor pressure [not required]

Osmosis and Osmotic Pressure

- https://www.youtube.com/watch?v=SrON0nEEWmo

- Osmosis: movement of solvent (usually water) across a semipermeable membrane
  - The membrane only allows certain molecules to pass through (e.g. water and coloring)
  - The membrane blocks other molecules (e.g. proteins and sugars inside the egg)
  - The “natural” direction: water flows from “more water” to “less water”
- Reverse osmosis [what is the application?]
  - The “unnatural” direction might also be achieved, but you have to apply an external pressure to force water to flow from “less water” to “more water”

Freezing-point Depression

- Drop of freezing point of the liquid (usually water) when solute is dissolved
- What is the application?

- Formula: $\Delta T_f = iK_f m$
  - ΔT_f: the difference of the melting point between the pure solvent and the solution
  - m: molality – typical way to express concentration in colligative property calculation

$$m = \frac{n(solute)}{m(solvent)} \quad \text{in the unit of mol kg}^{-1}$$

  - K_f: cryoscopic constant, which is only dependent on the solvent, but independent on the solute. For water, the value is 1.86 K kg mol^{-1}
  - i: van't Hoff factor – number of ion particles per formula unit of solute
    - Non-electrolyte (e.g. glucose, sucrose, etc.): i =1
    - Electrolyte: NaCl (i = 2); Li_2S (i = 3); K_3PO_4 (i = 4)

Boiling-point Elevation

- Raise of boiling point of the liquid (usually water) when solute is dissolved
- Formula: $\Delta T_b = iK_b m$
  - ΔT_b: the difference of the boiling point between the pure solvent and the solution
  - m: molality $$m = \frac{n(solute)}{m(solvent)}$$ in the unit of mol kg^{-1}
  - K_f: ebullioscopic constant, which is only dependent on the solvent, but independent on the solute. For water, the value is 0.512 K kg mol^{-1}
  - i: van't Hoff factor – number of ion particles per formula unit of solute
    - Non-electrolyte (e.g. glucose, sucrose, etc.): i =1
    - Electrolyte: NaCl (i = 2); Li_2S (i = 3); K_3PO_4 (i = 4)

Colloids 胶体

- A large group of mixtures intermediate between suspensions and solutions are called colloids.

- Colloids have particles larger in size than a true solution.

- Colloids are characterized by slow diffusion or no diffusion.

- Colloids share many properties with solutions — the particles in both are invisible without a microscope, do not settle on standing, and pass through filter paper.

Tyndall Effect

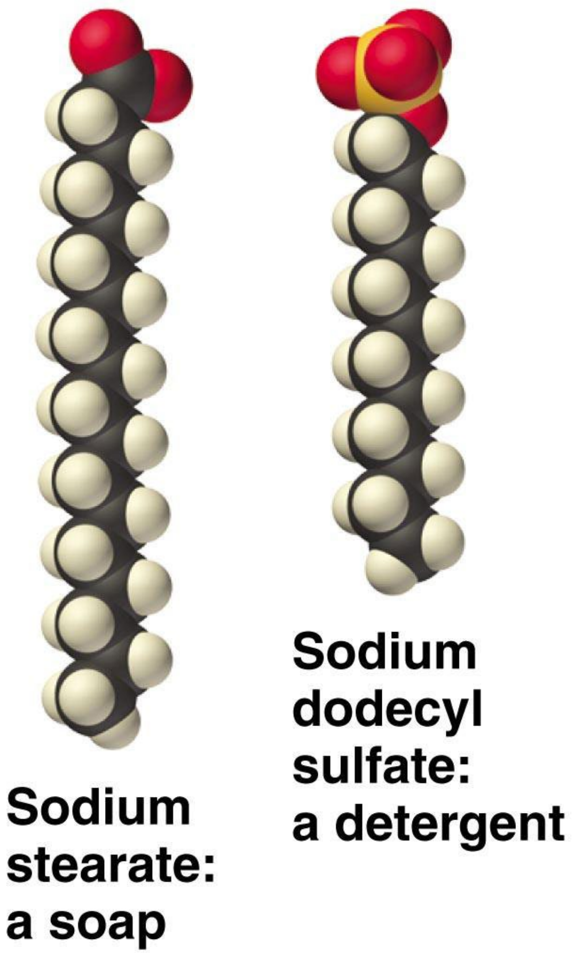
- Particle size: about 10 – 1000 nm
- Comparable with the wavelength of visible light 400 – 800 nm
- Light scattering occurs because of the comparable particle size

Types of Colloids

Colloid Type	Dispersed Substance	Dispersing Medium	Example(s)
Aerosol	Liquid	Gas	Fog
Aerosol	Solid	Gas	Smoke
Foam	Gas	Liquid	Whipped cream
Solid foam	Gas	Solid	Marshmallow
Emulsion	Liquid	Liquid	Milk
Solid emulsion	Liquid	Solid	Butter
Sol	Solid	Liquid	Paint; cell fluid
Solid sol	Solid	Solid	Opal

Emulsions

- **Emulsions are colloids formed by the dispersion of a hydrophobic liquid in water, thereby bringing two mutually insoluble liquids in close contact.**
- Various agents (surfactants) have been developed to stabilize emulsions, the most successful being molecules that combine a relatively long hydrophobic tail with a hydrophilic head.
- Examples of emulsifying agents include soaps and detergents.



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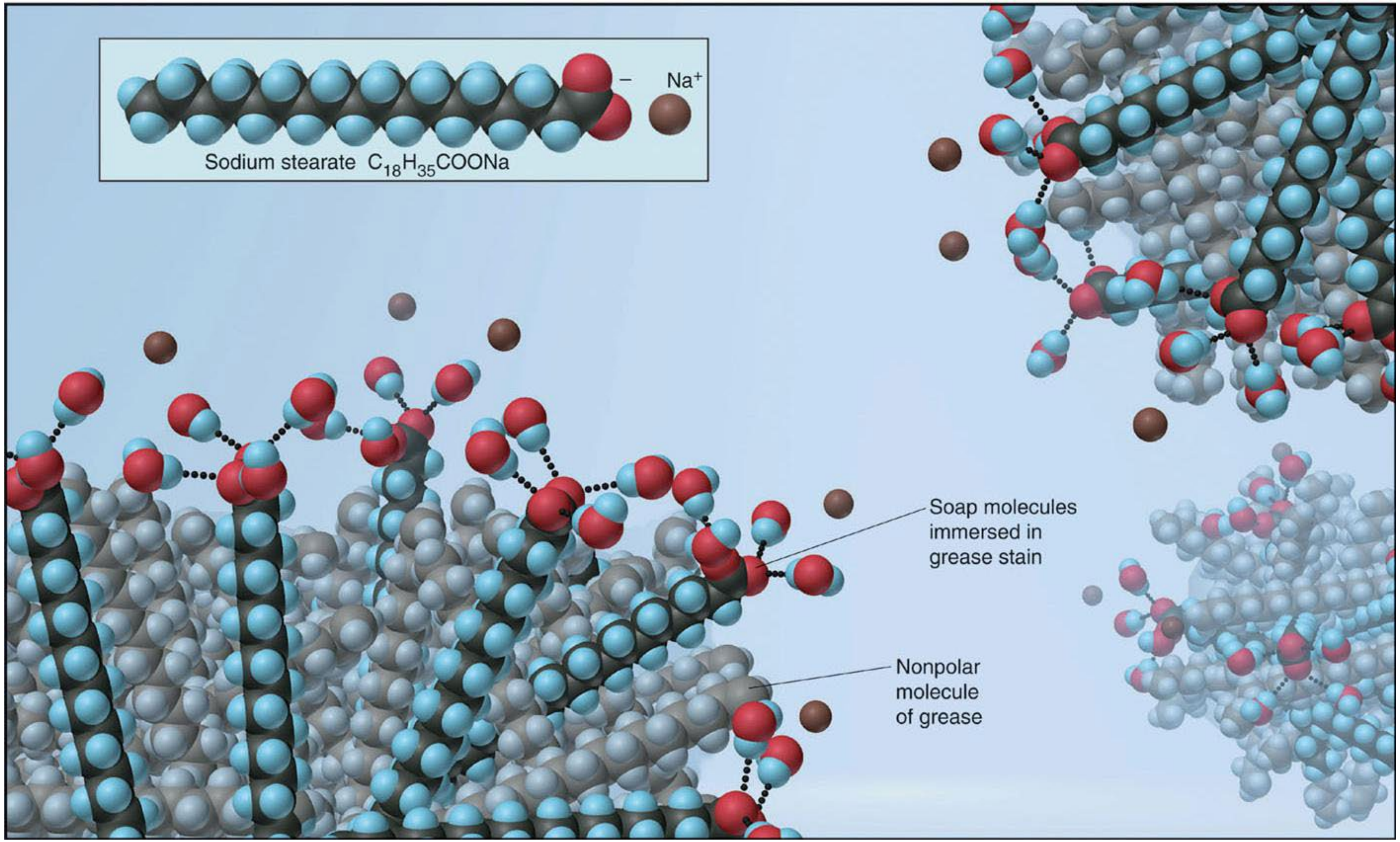
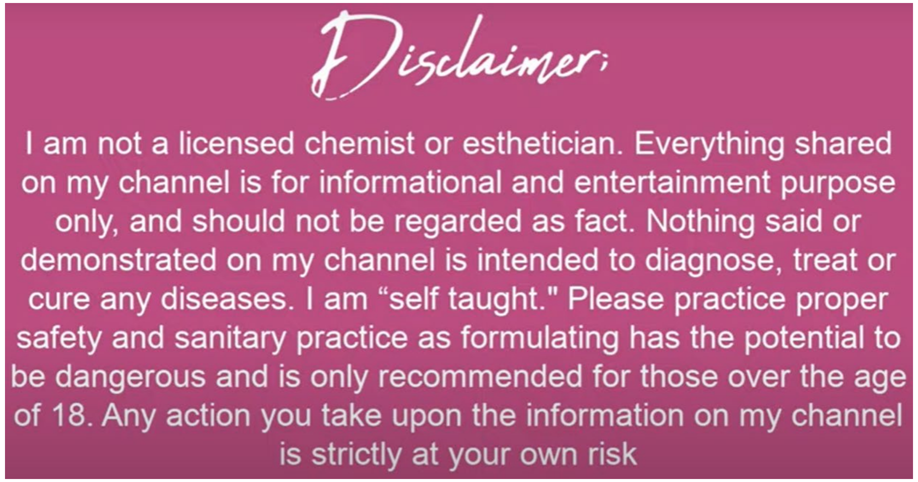


Na+
Sodium stearate C18H35COONa
Soap molecules immersed in grease stain
Nonpolar molecule of grease

How are they made?
cosmetic lotions, creams, moisturizers or hair conditioners

https://www.youtube.com/watch?v=aN6OZabVfG0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